

FaceTrack CRM — Technical Design Document

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1. System architecture



Strict one-way flow: FacePipeline → Gate → Scoring → DB; the LLM hangs off the side as a presentation-layer convenience.

2. Imaging pipeline · src/facetrack/cv_pipeline.py

- **Model:** MediaPipe Face Landmarker (Tasks API, vendored face_landmarker.task, 3.6 MB). 478 landmarks + a 4×4 facial-transformation matrix consumed downstream for pose.
- **Alignment + scale normalization:** one affine warp rotates around the eye-centre by the eye-line angle (indices 33 / 263) so the inter-pupillary line is horizontal, **and rescales the face so the anatomical width (landmark 234 ↔ 454) equals a fixed 512 px**. All five scoring metrics use fixed pixel-size kernels (15×15 black-hat, $\sigma = 1.4$ LoG, 3×3 Sobel), so without scale normalization the *same skin* photographed at a different distance or resolution scored differently — measured drift reached **+5 points on pore/wrinkle when the input was halved**. 512 px is the lowest-common-denominator scale: every photo the gate accepts (native face width ≥ 400 px) reaches it by *downscaling*, never by detail-fabricating upscaling. The same matrix is applied to the landmarks in homogeneous coordinates so ROI geometry stays in aligned-image space.
- **ROIs:** four anatomical regions (LEFT_CHEEK, RIGHT_CHEEK, FOREHEAD, CHIN) defined as **polygon paths over the aligned landmarks** (forehead = 18 indices, cheeks ≈ 10 each, chin ≈ 12), then rasterised to per-region binary masks. Scoring runs strictly **inside the mask** (scoring._ratio_inside / _stat_inside), so background pixels — hair, jewellery, clothing — can never contaminate the texture metrics. Polygons were chosen over the original axis-aligned bounding boxes once it became clear that a forehead bbox bleeds into the hairline on most face shapes.
- **Per-photo CLAHE** on LAB-L of each ROI normalises *within-photo* lighting; *cross-photo* normalisation is the gate's job, not the scorer's. When the gate's gray-card calibration fires, **the pipeline re-runs on the calibrated pixels before scoring**, so the stored photo and the stored score always agree (previously the score came from uncalibrated pixels while the calibrated photo was persisted — re-scoring the saved file would not have reproduced the saved number).

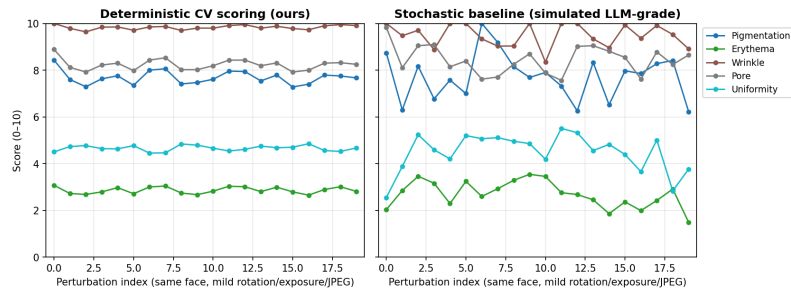
3. Model reliability & explainability · src/facetrack/scoring.py

Score	Formula	Why this choice
Pigmentation	MORPH_BLACKHAT(gray, 15×15) pixel ratio > 18, after a 3×3 Gaussian denoise	Black-hat highlights small dark structures — the morphological signature of melanin spots. The denoise step (same kernel the wrinkle metric already used) exists because the cutoff otherwise counts sensor/CLAHE noise, whose level varies with the input's downscale factor — measured as ~25 % raw drift between full- and half-resolution captures of the same face.
Erythema	mean of LAB.a* over the ROI mask	Standard clinical proxy; a* is luminance-independent after the gate calibrates colour.
Wrinkle	Sobel-magnitude > 30 pixel ratio (post-Gaussian)	Cheap, isotropic edge-density proxy for fine-line content.
Pore	LoG > 0.045 at $\sigma=1.4$, pixel ratio	Textbook isotropic-blob filter at pore-sized scale.
Uniformity (inv.)	std(LAB.L) mapped 0-10, inverted	Low variance $\Rightarrow$ uniform tone. The only metric where higher is better.

Every metric runs on an **effective mask**: the anatomical ROI polygon minus specular-glare pixels ($L^* > 235$) and deep-shadow pixels ($L^* < 20$), eroded 1 px so the artifact-to-skin transition ring is excluded too. Clinic downlights put a glare patch on most foreheads; that glare previously counted as "non-uniform tone" and its rim as "edges", and near-black hair strands counted as melanin spots. If exclusion would remove > 70 % of an ROI the scorer falls back to the full polygon — scoring a near-empty region is worse than scoring a glary one, and the gate's exposure check owns that failure mode.

Each raw measurement is linearly clamped against a published constant (PIGMENTATION_RAW_RANGE, etc.), calibrated on the five evenly-lit reference faces at the v2 normalization scale. **Re-calibrating to a clinic's distribution = editing one tuple, not retraining a model**. Formula changes bump SCORING_VERSION (now 2), which is **persisted on every visit row** — a longitudinal chart can annotate a version boundary, but it can never silently mix numbers produced by different formulas. Pre-v2 rows are backfilled to version 1 by a zero-downtime migration.

Reproducibility. Zero stochastic ops, zero LLM calls, zero network I/O in the scoring path. Same input $\rightarrow$ bit-identical output, enforced by tests/test_scoring_determinism.py (8 tests, including a no-input-mutation guard). Under mild perturbation (rotation $\leq 1.5^\circ$, exposure ± 4 %, JPEG re-encode q = 82-95; 20 trials on one face), scripts/reproducibility_evidence.py yields $\bar{\sigma}(\text{ours}) \approx 0.173$ vs $\bar{\sigma}(\text{stochastic baseline at } \sigma=1.0) \approx 0.686$ — a ~4× tighter band (per-metric: pigmentation 0.286, erythema 0.139, wrinkle 0.091, pore 0.232, uniformity 0.119; the wrinkle value partly reflects ceiling-clamp on this high-texture reference face — see BUILD_NOTES §4). An honesty note on the v1→v2 comparison: v1 reported $\bar{\sigma} \approx 0.074$, but two of its five metrics (wrinkle 0.000, pore 0.000) were *saturated at 10.0* on the reference face — a clamped score cannot vary, so part of that figure was range mis-calibration masquerading as robustness. v2's recalibrated ranges put scores mid-band where perturbation sensitivity is actually visible. The metric that matters for the product improved where it counts: **cross-resolution drift on the same face dropped from up to 5.5 points (v1) to ≤ 1.05 points (v2)** on the reference set, with undersampled inputs now rejected at the gate instead of silently mis-scored. The right panel uses a *simulated* baseline (no live Vision-LLM credit was available in the 48 hr window); the script is parameterised so swapping in a Claude vision call regenerates it without other changes.



Explainability. The score formula is the explanation. "Why is pigmentation 7.2?" is answered with a heatmap of the black-hat response — the same intermediate the score is computed from — surfaced directly above the score card (`visualization.py::metric_response_map`). The history page re-uses the same composer behind flat toggles (Streamlit forbids nested expanders), so the doctor ↔ patient comparison stays interactive across visits.

Scores-only contract — the structural anti-"thin-wrapper" guard. The Explainer Protocol accepts a `RegionScores` dataclass + a short patient-context string — **never pixels, never the image path, never the `QualityReport`**. The factory `get_explainer()` resolves Anthropic / Gemini / Mock by the `LLM_BACKEND` env var or auto-selects (Anthropic → Gemini → Mock) based on which keys are present; both real backends try/except to MockExplainer on SDK error, so the demo never hard-fails. Because the contract is *structural* (the Protocol's `explain()` signature has no `image` parameter), no code path lets the LLM synthesise a numeric score — even if a future contributor wanted it to. **Extensibility:** each scoring function is `bgr → float`; a new procedure-specific metric (volume change for filler, vascularity for rosacea) is one function plus one aggregator entry. No retraining, no schema migration. PRD §5's "next procedure" is a plug-in, not a rewrite.

4. Photo-Consistency Gate — the depth area · `consistency_gate.py`

Six checks gate every intake photo before scoring. Thresholds are calibrated on the reference faces in `data/test_images` (measured value bands quoted below); face-level checks run on the pipeline's scale-normalized aligned image — the same pixels the scorer sees.

- Pose.** Decompose MediaPipe's 4×4 facial-transformation matrix into yaw / pitch / roll (ZYX Euler). Reject if any axis exceeds `POSE_TOLERANCE_DEG = ±15°` in frontal mode; profile mode uses `PROFILE_YAW_MIN_DEG = 5°` for side captures. The threshold was relaxed from a DSLR-tuned $\pm 8^\circ$ once live webcam capture went in — natural laptop-camera tilt puts roll at -8° to -10° , so $\pm 8^\circ$ was unreachable without forcing unnatural, rigid intake photos (see `BUILD_NOTES`).
- Exposure — on the face crop.** Fraction of near-black (< 10) and near-white (> 245) pixels plus a mean-brightness band `[60, 210]`, measured on the **face bounding box** when landmarks exist (full-frame fallback otherwise). Full-frame measurement rejected well-lit faces in front of dark clinic walls and passed blown-out faces in front of mid-tone walls — the wall is not the patient. The near-black budget is looser on faces (6 % vs 2 % full-frame) because pupils, eyebrows and nostril shadows are legitimately near-black (measured 2.2–3.2 % on healthy reference faces); true underexposure still trips the mean floor.
- Sharpness — resolution-normalized + sampling floor.** Two-part: (a) native face width must be $\geq \text{MIN_NATIVE_FACE_WIDTH_PX} = 400$ — below that the photo does not carry the texture the metrics measure, and upscaling cannot fabricate it, so the honest answer is "請靠近鏡頭" rather than a fabricated score; (b) Laplacian variance on the face crop **resized to a fixed 256 px width**, so the threshold is device-independent (sharp reference faces measure 87–494 after normalization; `GaussianBlur(51, 25)` versions measure 1.9–3.7; threshold 40). The v1 unnormalized threshold had to be re-tuned from 80 to 30 when the capture device changed from DSLR to webcam — normalization removes that class of re-tuning permanently.
- Lighting uniformity (new in v2).** Relative left/right mean-brightness difference on the face crop, rejected above `LIGHTING_ASYMMETRY_MAX = 0.25`. A side-lit face passes every per-pixel exposure statistic, yet systematically biases the left-cheek vs right-cheek comparison and inflates uniformity/pigmentation on the shadow side — the exact cross-visit noise the gate exists to block. Clean separation on the reference set: evenly-lit faces measure 0.065–0.143, side-lit faces 0.31–0.52.
- Skin visibility (new in v2).** Per-ROI skin-pixel ratio via a YCrCb skin band, rejected when any ROI falls below `SKIN_RATIO_MIN = 0.35`, naming the occluded region ("偵測到遮擋：下巴..."). MediaPipe happily reports landmarks over a surgical mask or sunglasses, so before this check the scorer would obediently score fabric (LIMITATIONS §2 — now shipped rather than deferred). Reference faces measure ≥ 0.50 per ROI (worst case: a shadowed cheek); mask fabric and sunglasses measure 0.00. The YCrCb band is tuned for the Taiwan clinic population (Fitzpatrick II–IV); LIMITATIONS §4 tracks re-validation for V–VI.
- Color calibration.** Detect ArUco 5×5 markers (`DICT_5X5_50`); if present, sample the printed gray surround, compute per-channel gains — **clamped to [0.6, 1.8]** so a mis-sampled card (glare on the surround, marker on a colored sleeve) can never recolor the face harder than plausible clinic lighting, which would corrupt the erythema a^* metric worse than skipping calibration — and apply to the full image. If absent, **warn rather than hard-reject** — clinics adopt the calibration card progressively, so we degrade gracefully.

Failures yield actionable reasons (rendered to the receptionist in Traditional Chinese in the UI; the English equivalent here is "head turned 18.4° right; tolerance $\pm 15^\circ$; please face the camera"); the full `QualityReport` — all six checks with their raw measurements — is JSON-serialised onto the `Visit` row for audit. **Live-capture variant** (Session 2): a custom Streamlit component runs the same model **in the browser** via the Tasks Vision Web SDK and auto-captures only when pose / face-fill / stability all hold for `LIVE_CAPTURE_STABILITY_FRAMES = 10` consecutive frames — converting the gate from "reject after the fact" to "guide in real time". Face-fill is bounded to `[0.35, 0.75]` of frame width so pore / wrinkle metrics have enough pixels without clipping the chin (this floor also guarantees live captures clear the 400 px native-face-width gate). **The full six-check server-side gate still re-runs on the captured frame** — the browser HUD is a UX accelerator, not a security boundary.

5. Workflow integration

Three-actor flow: receptionist captures (live or upload) → server-side gate runs → on pass, 5 metrics × 4 ROIs + LLM draft are produced → physician (sitting next to the receptionist) reviews the visit-history page with the ROI overlay toggle, edits the treatment-plan draft, saves → longitudinal page updates. The patient sees the heatmap + explanation on-screen during the consult.

Streamlit pages (6, sidebar nav defined in `app.py::PAGE_LABELS`):

Key	Label (UI)	Owns
patients	Patient management	CRUD + soft-delete + restore
intake	New visit	Live face-mesh capture + upload fallback + gate + scoring + LLM draft + save
history	Visit history	Per-visit timeline + ROI overlay toggles + CLAHE thumbnails
treatment	Treatment plan	Editable plan tied to the latest visit
overview	Longitudinal tracking	Radar chart (current) + line chart (all visits × 5 metrics)
settings	Settings	Gate thresholds, LLM backend status, audit toggles

Stack rationale. Streamlit (single-file UI, one-click Cloud deploy — the right tool for a 48 hr CV-pipeline demo; React + FastAPI is a 1-week refactor, deferred). Where Streamlit was too restrictive (no nested expanders, no client-side ML) we dropped to a `declare_component` widget for live capture — the Python side just consumes `{front, left?, right?}` and the rest of the pipeline is unchanged. `SQLModel` (same `BaseModel` API as the rest of the codebase; single `SQLite` file ships in the repo). Migrations are zero-downtime `ALTER TABLE ADD COLUMN`. State lives on disk — no Redis, no workers, no queues — the entire pipeline runs synchronously in the request thread because the slowest stage (`MediaPipe`) is < 200 ms on Apple Silicon. History-view hot paths use `@st.cache_data` keyed on `(path, mtime)` so toggling the ROI overlay does not re-run `MediaPipe`.

6. Stack, cost, latency · `scripts/benchmark.py` (M4 Pro, 3 faces × 5 runs)

Stage	p50	p95
Alignment + scale-norm + ROI extraction (<code>MediaPipe</code> Tasks)	7.7 ms	8.1 ms
Consistency Gate (6 checks + <code>ArUco</code>)	6.3 ms	6.6 ms
Scoring (5 metrics × 4 ROIs, effective-mask exclusion)	7.6 ms	8.1 ms
End-to-end (excl. LLM)	21.6 ms	22.5 ms
Explainer (Claude Sonnet 4.6, network)	~1.5 s	~2.5 s

Python 3.11 (`mediapipe` 0.10 has spotty 3.12 wheels on macOS arm64). uv-managed deps, ruff-formatted, pytest-tested (97 tests / 10 files: 92 fast + 5 opt-in ground-truth benchmarks). `anthropic` + `google-genai` SDKs are optional; absent both, the app runs against `MockExplainer` and the loop still works end-to-end. Per-visit LLM cost at Sonnet 4.6 pricing ≈ **\$0.0048 / visit** (~600 in + ~200 out tokens); 1 000 visits / month ≈ **\$5 / month** — a rounding error vs. the human time saved drafting the treatment note.

7. Limitations · full catalogue in [docs/LIMITATIONS.md](#)

Scoring ranges are calibrated on 5 reference photos (a real pilot would re-fit on ~200 per Fitzpatrick type). No identity confirmation on intake (Phase 2: face-embedding vs. the patient's first-visit photo). `ArUco` card adoption is voluntary (Phase 2: required-marker clinic setting). No HIPAA / PIPL story — photos sit in clear on disk (Phase 2: at-rest encryption + a deletion API). Skin-surface visibility is now partially validated: the v2 skin-visibility check catches masks / sunglasses / hair occlusion, but **heavy makeup and smartphone beauty filters still pass every gate check** and produce misleading scores — and the `YCrCb` skin band needs re-validation for Fitzpatrick V–VI. Ground-truth grading of the gate and the five scores against independent public datasets (`FFHQ-Wrinkle`, `ACNE04`, `SCIN`) is in `docs/VALIDATION.md`. **What was reused vs. built** is documented in `docs/BUILD_NOTES.md` per the brief.